

Table of contents page

Executive Summary	2
Systems and Procedures.....	3
Results.....	4
Conclusion	5
Appendix 1.....	6
Appendix 2.....	6
Appendix 3.....	7
References.....	8

Executive Summary

Production of cyclohexane was simulated using Excel via the hydrogenation of benzene in a bubble-column reactor. Preparation of a material balance along with the operating data and assumptions were established. The tool, CHE 433 Calculator, was given to solve the problems. Initial parameters were given such as fresh feeds, conversion, selectivity, and molar ratios to carry out the exercise. Changes in VBA code was done in order to fit the parameters and streams of cyclohexane production. Once the code was fixed, the simulation was started for it to converge from the tear stream and give us the results. If the simulation results matched the operating data, the simulation was successful, and the data was recorded. The converged simulation results were used to record the estimated overall conversion, selectivity, mass flow rate of product, and purity. Additionally, we had to comment on the purity of the cyclohexane to see if it meets market specifications and to estimate the fixed capital investment for a plant using 2019 dollars. Finally, we were to propose a method to remove the liquid recycle for production of 200 KMTA for a reduction in raw material costs.

Systems and Procedures

Our first step was to input the given temperatures and pressures for the fresh feed streams and for our operating data on the reactor itself. Knowing that we have the conversion and selectivity per pass, we were able to solve for δ_1 and δ_2 using the fact that $\delta_1 + \delta_2 = Conv_{pass} * n^{\circ} C_6H_6$ and $\frac{\delta_1}{\delta_1 + \delta_2} = Sel_{pass}$. The known values were inserted into our code to solve for the conversion and selectivity of the reactor. Now, looking at our PFD we shift our focus onto the splitters where we have HP and LP columns. Knowing the temperatures and pressure going into the reactor, we can use the z_i (composition) and the k-values given to solve for ' α ' using the Rachford-Rice equation: $f(\alpha) = \frac{z_i(1-k_i)}{1+\alpha(k_i-1)} = 0$. Once we get our alpha, we know $\alpha = \frac{v}{f}$. Therefore, we can use this equation to solve for our vapor flowrate and then use the fact that $F - V = L$, to solve for our liquid flowrate. Using these values, we can solve for y_i and x_i coming out of the HP using these relations: $X_i = \frac{z_i}{1+\alpha(K_i-1)}$ $y_i = \frac{z_i K_i}{1+\alpha(K_i-1)}$. Next, we shifted our focus on the LP. Knowing the liquid flowrate out the HP is S3, we multiply our bottoms recycle factor by S3 to get S7 which is our liquid recycle stream. Next we can see according to our PFD, that one minus our bottoms recycle factor multiplied by S3 will give us S8. We can solve for S10 and S11. Furthermore, we know the vapor stream coming out the HP is S4 so we can multiply S4 by our purge fraction which gave us S6. Inputting this into our code we make sure we added the parameter that our hydrogen to benzene ratio in our top recycle stream is about 3.3. We also inputted into our code the parameter that our liquid recycle stream cannot be greater than 45% of the benzene fresh feed stream. We updated the tear stream and adjusted the original codes cells for the streams, so when we run our code, all streams can converge except the two fresh feed streams which would remain the same.

Results

As we saw in our results, we were able to input all our operating data and obtain a flowsheet convergence from the tear stream using the fixed VBA code. Once convergence was obtained we were able to obtain the overall conversion and selectivity of benzene. Our overall conversion was 98.40 mol% and our overall selectivity was 99.7 mol%. Next, we obtained the mass flow rate of the cyclohexane and the purity of this product. Our mass flowrate was 36.18 KMTA while our purity of our product was 96.58 wt%. This purity is a bit lower than market specifications. According to science direct, "The purity level required for the use of cyclohexane, especially for its oxidation, is higher than 99%". Looking at our FCI formula of $FCI = aNF^b$, we know the plant capacity is 200 KMTA which is our 'F', and 'N' is the number of units in our process and using regression economic analysis to find 'a' and 'b'. Plugging our numbers in we get 700.65 MM(USD) as our FCI estimate for this plant. Our method for removing the liquid recycle stream would be to increase our purge stream in order to decrease the impurities going back into the recycle stream. Doing this, we eliminate our liquid recycle stream and increase the efficiency of the 200 KMTA manufacturer plant.

Conclusion

In conclusion we see how the production of cyclohexane all comes together through many different streams via VBA coding and our CHE 433 calculator as our tool to solve the problem and obtain the convergence. Doing this, we had our simulation match our operating data and thus give us the results for the overall conversion and selectivity of benzene. We also obtained the mass flowrate of our product and the purity of the cyclohexane product and compared It to real life market specifications of the purity of cyclohexane. We got to understand how to calculate our fixed capital investment using the equation mentioned above and seeing how the capacity of our plant and number of units in our process affects out FCI. We were able to propose a method to remove the liquid recycle stream which would help reduce the cost of our raw material and improve the efficiency of the 200 KMTA manufacturing plant.

Appendix 1

$$\delta_1 + \delta_2 = Conv_{pass} * n^{\circ} C_6 H_6$$

$$\frac{\delta_1}{\delta_1 + \delta_2} = Sel_{pass}$$

$$k_i = \frac{y_i}{x_i}$$

$$\alpha = \frac{v}{f}$$

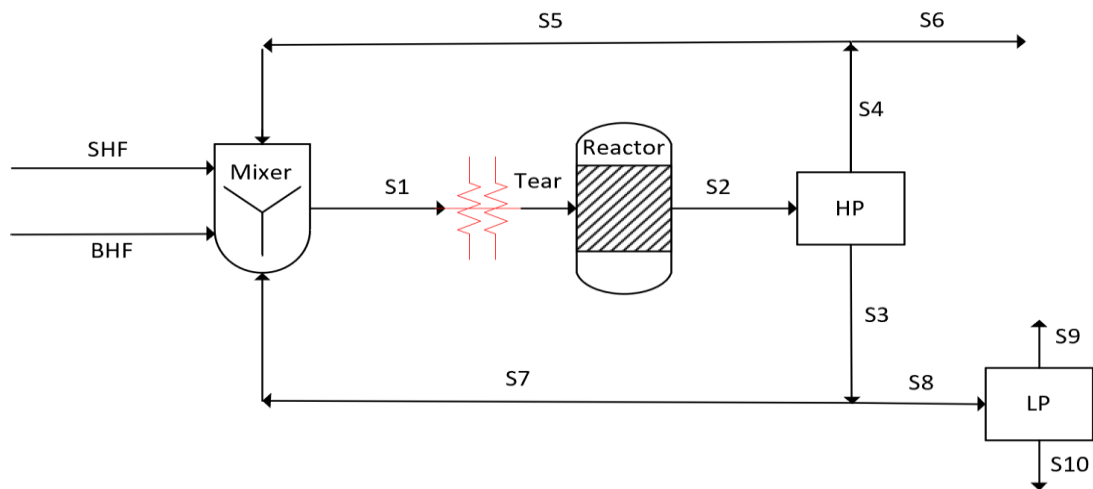
$$\frac{Z_i(1-k_i)}{1 + \alpha(k_i - 1)} = 0$$

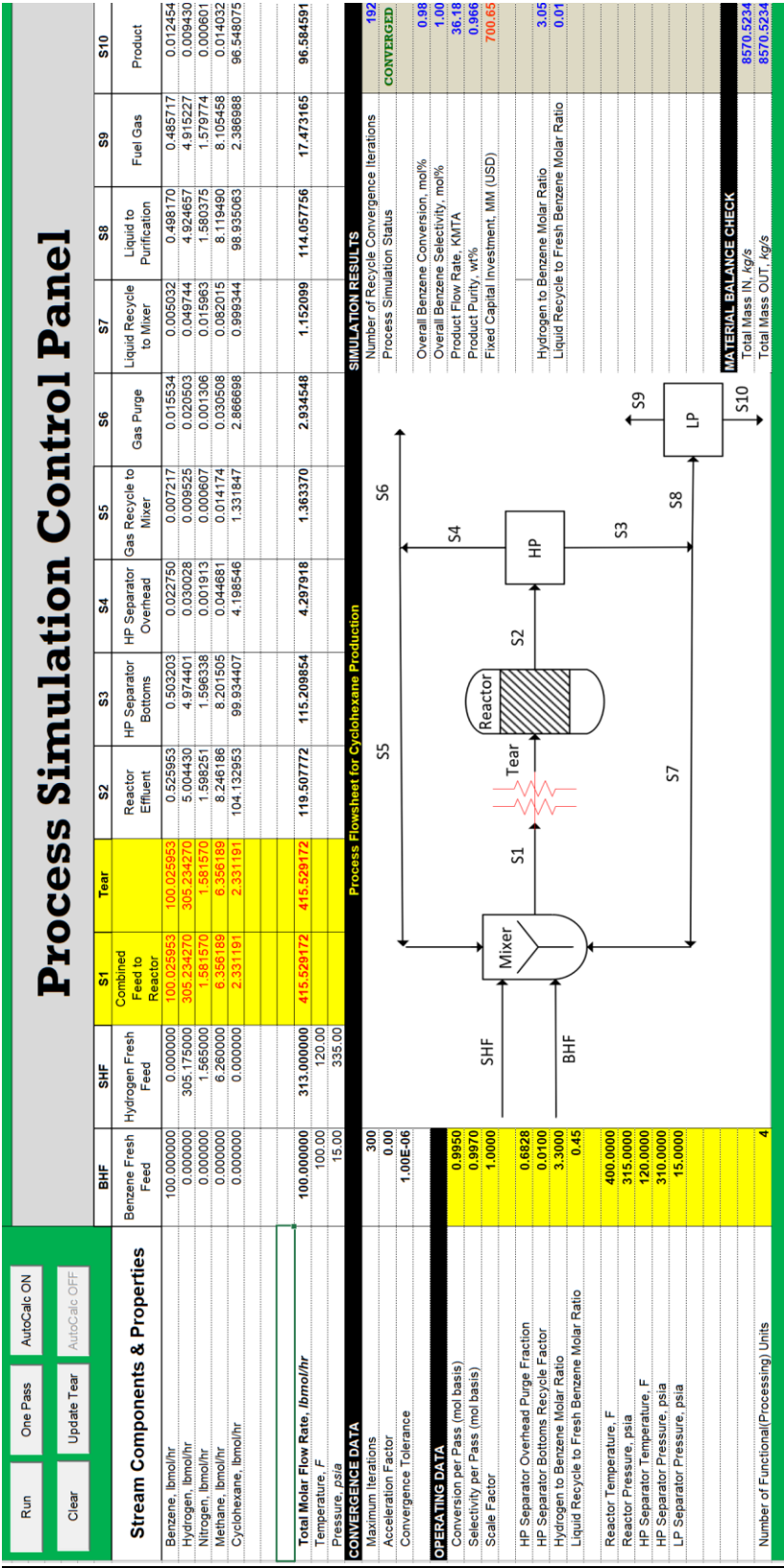
$$F - V = L$$

$$X_i = \frac{Z_i}{1 + \alpha(K_i - 1)} y_i = \frac{Z_i k_i}{1 + \alpha(K_i - 1)}.$$

Appendix 2

PFD of Cyclohexane Production





References

- “Cyclohexane.” *Cyclohexane - an Overview | ScienceDirect Topics*,
www.sciencedirect.com/topics/chemical-engineering/cyclohexane#targetText=Cyclohexane%20is%20an%20essential%20intermediate,by%20the%20benzene%20hydrogenation%20technique.
- Zhang, Chi. “Estimate the Capital Cost of Shale-Gas Monitization Projects.” *Energy*, learn-us-east-1-prod-fleet01-xythos.s3.us-east-1.amazonaws.com/5cc71db6522fe/6306281?response-content-disposition=inline%3B%20filename%2A%3DUTF-8%27%27CAPEX%2520Estimation%25281%2529.pdf&response-content-type=application%2Fpdf&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20190922T150305Z&X-Amz-SignedHeaders=host&X-Amz-Expires=21600&X-Amz-Credential=AKIAIBGJ7RCS23L3LEJQ%2F20190922%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Signature=d3006bb1b7b4aa5fb7b54dbba54f2427d86e8d9e28548ad9caf5f6da34cb4f2c.